• **DRIVEABILITY—SPARK KNOCK—DURING ACCELERATION—VEHICLES EQUIPPED WITH 3.0L ENGINE BUILT THROUGH 2/21/2000** • **NOISE—SPARK KNOCK—DURING ACCELERATION—VEHICLES EQUIPPED WITH 3.0L ENGINE BUILT THROUGH 2/21/2000**	**Article No. 00-24-8**

FORD: 1998-2000 RANGER

This article is being republished in its entirety to update the Service Procedure and the Parts required.

ISSUE

Some vehicles equipped with a 3.0L engine may exhibit spark knock (ping) under acceleration. This may be caused by the ignition coil and the Powertrain Control Module (PCM).

ACTION

Replace the ignition coil and PCM assembly. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

1. Verify spark knock by road testing vehicle.
2. Verify that the spark knock is in the combustion chamber by completely removing the fuel from the fuel tank and adding 2 gallons of 94 octane fuel.
3. Road test vehicle, check for spark knock, and then return the original fuel to the tank.
4. If the spark knock was still present during the road test, this TSB does not apply to your condition. Pursue other normal mechanical noise diagnostics and repairs.
5. If spark knock was eliminated during the road test, unplug the ignition wires from the coil and read the stamped black 5-digit code on the label (four numbers and one letter).
6. If the first four numbers of the code fall in the numerical range from 7001 to 9213, replace coil with revised Coil (F5SZ-12029-A). If the service coil label code falls in this range, obtain another service coil and again check the code on the label of the new service coil. Return coils that fall within the range for a refund per your part return policy. Only use the service coil if the first four numbers of the code DO NOT fall within the numerical range 7001 to 9213.
7. Replace the PCM assembly with a revised PCM assembly as per chart below. Refer to the appropriate Ranger Workshop Manual for removal and installation procedures.

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PCM APPLICATION				
Application	**Old Part Number (-12A650-)**	**Old Tear Tag**	**New Tear Tag**	**New Part Number (12A650-)**
GAS VEHICLES				
1998 M/T 4X2 49S/ALT/CAN	F87F-XAMR, BMR	EFA0, 1	WSN0	F8PZ-12A650-EPA
Mexico (Service)	WU7A-AAMR	ERA0, 1	RPZ1	WU7Z-12A650-AB
1998 M/T 4X2 California	F87F-AAAMR, BMR	HND0, 1	YPK0	F8PZ-12A650-ETA
1998 A/T 4X2 49S/ALT/CAN	F87F-BNAMR	BPE0	KSK0	F8PZ-12A650-EKA
Mexico (Service)	WU7A-BAMR	FPE0	RNV1	WU7Z-12A650-BB
1998 A/T 4X2 California	F87F-BSAMR	EMR0	TXW0	F8PZ-12A650-ENA
1998 A/T 4X4 49S/ALT/CAN	F87F-BPAMR	CVV0	WLB0	F8PZ-12A650-ELA
1998 A/T 4X4 California	F87F-BRAMR	DGG0	JPG0	F8PZ-12A650-EMA
1998 M/T 4X4 49S/ALT/CAN	F87F-YAMR, BMR	FLQ0, 1	YUY0	F8PZ-12A650-ERA
1998 M/T 4X4 California	F87F-ZAMR, BMR	GTA0, 1	UKF0	F8PZ-12A650-ESA
1999 A/T 4X2 CAN	XL5F-FAMR, BMR	JAA0, 1	YVR0	XU2Z-12A650-AZA
Mexico (Service)	XU2F-AZAMR	JAA0, 1	RYK0	XU7Z-12A650-BAM
1999 A/T 4X4 CAN/MEX	XL5F-GAMR, BMR	GRR0, 1	HXG0	XU2Z-12A650-BAA
1999 1/4-2000 M/T 4X2 Trailhead/Troy Lee CAN/MEX	XU3F-ABAMR	ZZP0	HFD0	YL5Z-12A650-AHA
1999-2000 M/T 4X4 CAN/MEX	XL5F-AHAMR, BMR	PDR0, 1	TEP0	YL5Z-12A650-AGA
1999 1/4-2000 A/T 4X2 Trailhead/Troy Lee CAN/MEX	XU3F-ACA, B YL5F-RAMR	RFH0, 1 MKG0	GXK0	YU7Z-12A650-TA
1999 1/4-2000 A/T 4X4 CAN/MEX	XL5F-AUA, B YL5F-AKAMR	MTD0, 1 ZRC0	XWD0	XU7Z-12A650-SA
1999-2000 M/T 4X2 CAN	XL5F-AGAMR, BMR	MBJ0, 1	NXV0	YL5Z-12A650-AFA
Mexico (Service)	XL5F-AGBMR	MBJ0, 1	RXC0	XU7Z-12A650-CA
1999 1/4-2000 A/T 4X2 CAN	XL5F-ATAMR, BMR YL5F-HJAMR	LLF0, 1 FLU0	YYI0	YU7Z-12A650-RA
Mexico (Service)	XU7A-DAMR	RDV0	RDV1	XU7Z-12A650-DB
FLEX FUEL VEHICLES				
1999 A/T 4X2 FFV/California	XL5F-KAMR, KCMR XU2F-HA MR	KDM0, 2 NCH0	WTV0	XU2Z-12A650-BCA
1999 1/4-2000 A/T 4X2 FFV/CAL	XL5F-KDMR, KEMR, KFMR, KGMR - YL5F-VAMR	KDM3,4,5,6 KGZ0	OER0	YU7Z-12A650-MA
1999 A/T 4X2 FFV/FED/ALT/CAN	XL5F-HAMR, HCMR XU2F-FAMR	HCL0,2 MJH0	HEH0	XU2Z-12A650-BBA
1999 1/4-2000 A/T 4X2 FFV/FED/ALT/CAL	XL5F-HDMR, HEMR, HFMR, HGMR - YL5F-SAMR	HCL3,4,5,6 QGF0	XCW0	YU7Z-12A650-UA
1999 A/T 4X4 FFV/FED/ALT/CAN	XL5F-JAMR, JCMR, XU2F-GAMR	JTQ0,2 RGJ0	CPK0	XU2Z-12A650-BDA
1999 1/4-2000 A/T 4X4 FFV/FED/ALT/CAL	XL5F-JDMR, EMR, FMR, GMR YL5F-TAMR	JTQ3,4,5,6 TIH0	XIF0	YU7Z-12A650-KA
1999 A/T 4X4 FFV/California	XL5F-LAMR,CMR XU2F-JAMR	LYC0,2 YRB0	WRJ1	XU2Z-12A650-BEB
1999 1/4-2000 A/T 4X4 FFV/CAL	XL5F-LDMR, EMR, FMR, GMR YL5F-XAMR	LYC3,4,5,6 PUL0	WAJ0	YU7Z-12A650-NA
1999-2000 M/T 4X2 FFV/FED/ALT/CAN	XL5F-MAMR, CMR, DMR, EMR, FMR, GMR	MXA0, 2, 3, 4, 5, 6	FHZ0	YL5Z-12A650-ZA
1999-2000 M/T 4X2 FFV/California	XL5F-YAMR, CMR, DMR, EMR, FMR, GMR	YBJ0, 2, 3, 4, 5, 6	EZU0	YL5Z-12A650-ACA

Application	Old Part Number (-12A650-)	Old Tear Tag	New Tear Tag	New Part Number (12A650-)
1999-2000 M/T 4X4 FFV/FED/ALT/CAN	XL5F-NAMR, CMR, DMR, EMR, FMR, GMR	NTT0, 2, 3, 4, 5, 6	NEE0	YL5Z-12A650-AAA
1999-2000 M/T 4X4 FFV/California	XL5F-PAMR, CMR, DMR, EMR, FMR, GMR	PDL0, 2, 3, 4, 5, 6	ZQY0	YL5Z-12A650-ADA
1999 1/4-2000 M/T 4X2 Trailhead/Troy Lee FFV/FED/ALT/CAN	XU3F-ADAMR, BMR	PFK0,1	FRP0	YL5Z-12A650-ABA
1999 1/4-2000 M/T 4X2 Trailhead/Troy Lee FFV/California	XU3F-AFAMR, BMR	MGG1	JRK0	YL5Z-12A650-AEA
1999 1/4-2000 A/T 4X2 Trailhead/Troy Lee FFV/FED/ALT/CAN	XU3F-AEAMR, BMR, CMR, YL5F-UAMR	GJM0,1,2 LRW0	DCH0	YU7Z-12A650-LA
1999 1/4-2000 A/T 4X2 Trailhead/Troy Lee FFV/California	XU3F-AGAMR, BMR, CMR, YL5F-YAMR	CUG0,1,2 PZT0	OIV0	YU7Z-12A650-PA

8. After PCM and coil replacement, road test to verify repair.

NOTE
ENGINE SPARK KNOCK WILL BE SIGNIFICANTLY REDUCED AFTER PCM REPLACEMENT. DO NOT BE CONCERNED IF THE ENGINE EXHIBITS SLIGHT SPARK ENGINE KNOCK (PING) UNDER CERTAIN DRIVING CONDITIONS AFTER PCM REPLACEMENT. THIS IS CONSIDERED A NORMAL CHARACTERISTIC FOR THIS PARTICULAR APPLICATION. REFERENCE YOUR APPROPRIATE OWNER'S MANUAL FOR FURTHER INFORMATION.

NOTE
IF THE VEHICLE IS PASSIVE ANTI-THEFT SYSTEM (PATS) EQUIPPED, REFER TO THE APPROPRIATE RANGER WORKSHOP MANUAL FOR PROPER PATS PCM INSTALLATION. ALL KEYS MUST BE PROGRAMMED TO START THE SUBJECT VEHICLE. VERIFY ALL OF THE CUSTOMER KEYS WILL START VEHICLE PRIOR TO RELEASING THE VEHICLE.

9. If spark knock (ping) is still slightly objectionable, you "may" proceed to perform the octane adjust.

   a. Obtain New Generation Star (NGS) Tester to perform engine octane adjust.

   b. Use NGS Service Card and Cable 4.2 level or higher and select:
      - Service Bay Functions
      - Powertrain Control Module
      - EEC V Configure
      - Press trigger to continue on emission compliance authorization screen
      - Highlight octane adjust column and press trigger
      - Rotate dial to change ignition timing from "no retard" to "retard" and then press trigger
      - Highlight the fuel octane and press trigger
      - Rotate the dial to set to 87 octane and press trigger
      - Press start button - a message stating "Change has been successful" will appear

NOTE
FAILURE TO PRESS THE START BUTTON WILL RESULT IN CHANGES IN OCTANE ADJUSTMENT NOT BEING ACCEPTED.

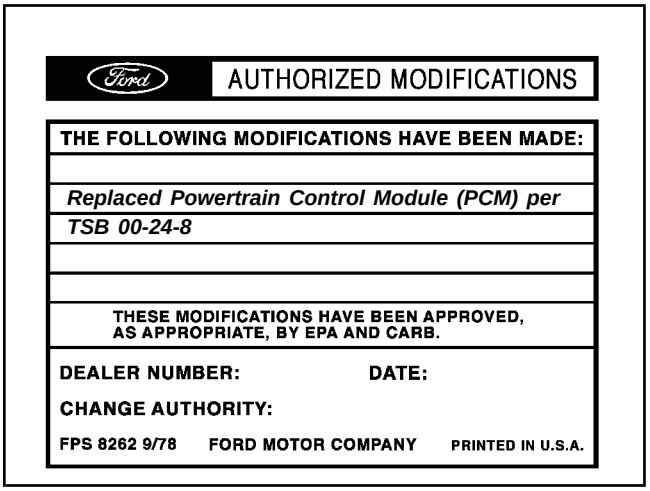
   - Exit out of Service Bay Functions

10. After octane adjust, road test and verify repair.

NOTE
ENGINE PERFORMANCE MAY BE SLIGHTLY REDUCED. OCTANE ADJUST AND TIMING RETARD CAN BE PERFORMED A MAXIMUM OF TWO TIMES WITHOUT REPROGRAMMING THE

PCM. IF YOU NEED TO PERFORM THIS PROCEDURE MORE THAN TWO TIMES IT WILL BE NECESSARY TO REPROGRAM THE PCM.

Obtain an Authorized Modifications Decal (FPS 8262 - obtainable through DOES II, 25/pkg) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



PART NUMBER	PART NAME
F5SZ-12029-A	Coil Assembly
F8PZ-12A650-EKA	Powertrain Control Module (PCM)
F8PZ-12A650-ELA	Powertrain Control Module (PCM)
F8PZ-12A650-EMA	Powertrain Control Module (PCM)
F8PZ-12A650-ENA	Powertrain Control Module (PCM)
F8PZ-12A650-EPA	Powertrain Control Module (PCM)
F8PZ-12A650-ERA	Powertrain Control Module (PCM)
F8PZ-12A650-ESA	Powertrain Control Module (PCM)
F8PZ-12A650-ETA	Powertrain Control Module (PCM)
WU7Z-12A650-AB	Powertrain Control Module (PCM)
WU7Z-12A650-BB	Powertrain Control Module (PCM)
XU2Z-12A650-AZA	Powertrain Control Module (PCM)
XU2Z-12A650-BAA	Powertrain Control Module (PCM)

(Cont'd.)

PART NUMBER	PART NAME
XU2Z-12A650-BBA	Powertrain Control Module (PCM)
XU2Z-12A650-BCA	Powertrain Control Module (PCM)
XU2Z-12A650-BDA	Powertrain Control Module (PCM)
XU2Z-12A650-BEB	Powertrain Control Module (PCM)
XU7Z-12A650-BAM	Powertrain Control Module (PCM)
XU7Z-12A650-CA	Powertrain Control Module (PCM)
XU7Z-12A650-DB	Powertrain Control Module (PCM)
YL5Z-12A650-AAA	Powertrain Control Module (PCM)
YL5Z-12A650-ABA	Powertrain Control Module (PCM)
YL5Z-12A650-ACA	Powertrain Control Module (PCM)
YL5Z-12A650-ADA	Powertrain Control Module (PCM)
YL5Z-12A650-AEA	Powertrain Control Module (PCM)
YL5Z-12A650-AFA	Powertrain Control Module (PCM)
YL5Z-12A650-AGA	Powertrain Control Module (PCM)
YL5Z-12A650-AHA	Powertrain Control Module (PCM)
YL5Z-12A650-ZA	Powertrain Control Module (PCM)
YU7Z-12A650-KA	Powertrain Control Module (PCM)
YU7Z-12A650-LA	Powertrain Control Module (PCM)
YU7Z-12A650-MA	Powertrain Control Module (PCM)
YU7Z-12A650-NA	Powertrain Control Module (PCM)
YU7Z-12A650-PA	Powertrain Control Module (PCM)
YU7Z-12A650-RA	Powertrain Control Module (PCM)
YU7Z-12A650-SA	Powertrain Control Module (PCM)
YU7Z-12A650-TA	Powertrain Control Module (PCM)
YU7Z-12A650-UA	Powertrain Control Module (PCM)

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-7-3
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
002408A	Replace Ignition Coil And Powertrain Control Module (PCM) With Or Without Passive Anti-Theft System (PATS)	2.2 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
12A650	42

OASIS CODES: 615000, 615500